

Quantum Gravitation

Lecture 6

Propagators and Medium Causality

- Degrees of freedom
- Quantization
- Spectrum and response
- Consistency checks
- Example

■ Degrees of freedom

We separate propagating polarizations from constrained components before quantization.

$$(\partial_t^2 - c_m^2 \nabla^2 + m_g^2 c_m^4 / \hbar^2) G_R = \delta^4(x)$$

■ Quantization

Canonical normalization is fixed before creation and annihilation operators are introduced.

$$D_{\mu\nu,\rho\sigma} = \Pi_{\mu\nu,\rho\sigma}^{(2)} / (k^2 - m_g^2 + i\varepsilon)$$

■ Spectrum and response

Poles, residues, and group velocity are read from the same quadratic operator.

$$S_{obs} = K_{det} * G_R * J$$

■ Consistency checks

Hermiticity, positive norm, source conservation, and the low-energy limit are checked explicitly.

■ Example

A short numerical estimate fixes the scale of the effect before a full fit is attempted.

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Reading: Seo, ch. 6; Cole, ch. 7; Detector Kernel Registry